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To know the number of molecules in the range of speed between 50 m s^{-1} and 60 m s^{-1}, we need to integrate $\int_{50}^{60} 4\pi N\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}} dv =$ $= N(50 \text{ to } 60\ ms^{-1})$. In general the number of molecules within the range of speed v and v+dv is given by $\int_{v}^{v+dv} 4\pi N\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}} dv = N(v \text{ to } v+dv)$.

The exact integration is beyond the scope of the book. But we can infer the behavior of gas molecules from the graph.

(i) The area under the graph will give the total number of gas molecules in the system

(ii) Figure 9.4 shows the speed distribution graph for two different temperatures. As temperature increases, the peak of the curve is shifted to the right. It implies that the average speed of each molecule will increase. But the area under each graph is same since it represents the total number of gas molecules.

Note

Interestingly once the gas molecule attains equilibrium, the number of molecules in the given range of speeds are fixed. For example if a molecule initially moving with speed 12 m s^{-1}, collides with some other molecule and changes its speed to 9 m s^{-1}, then the other molecule initially moving with different speed reaches the speed 12 m s^{-1} due to another collision. So in general once the gas molecules attain equilibrium, the number of molecules that lie in the range of v to v+dv is always fixed.

9.3 DEGREES OF FREEDOM

9.3.1 Definition

The minimum number of independent coordinates needed to specify the position and configuration of a thermo-dynamical system in space is called the degree of freedom of the system.

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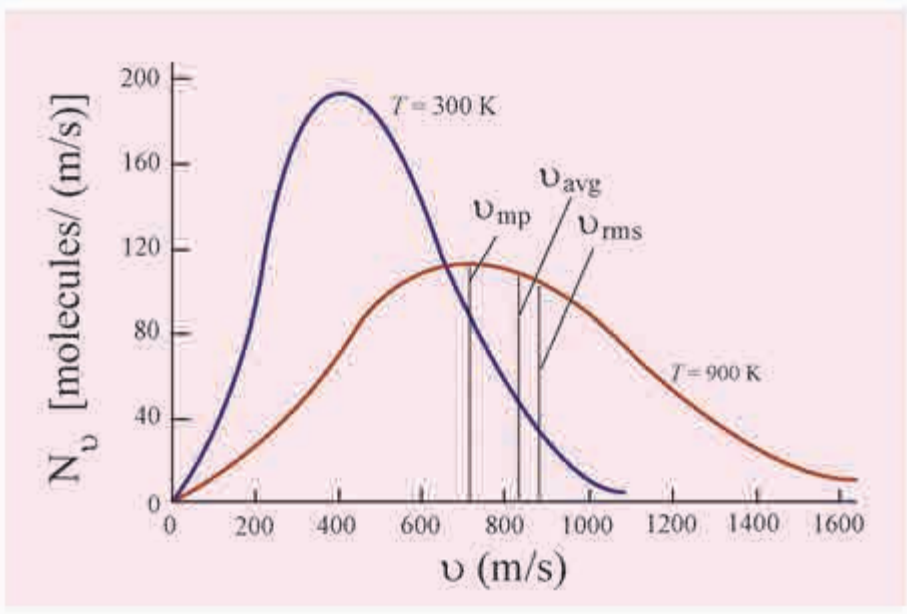


Figure 9.4 Maxwell distribution graph for two different temperatures